

SCRAMBLEBOX: An Arduino Based Word Jumble Game Console

MINIPROJECT REPORT

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to

the APJ Abdul Kalam Technological University

in partial fulfillment of the requirements for the award of the Degree

of

Bachelor of Technology

in

Electronics and Communication Engineering



**DEPARTMENT OF ELECTRONICS AND
COMMUNICATION ENGINEERING**

**SREE CHITRA THIRUNAL COLLEGE OF
ENGINEERING, THIRUVANANTHAPURAM**

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**DEPARTMENT OF ELECTRONICS AND
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ENGINEERING, THIRUVANANTHAPURAM**

2024 - 2025



CERTIFICATE

This is to certify that the report entitled

“SCRAMBLEBOX: An Arduino-Based Word Jumble Game Console” submitted by,
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to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for
the award of the Degree of Bachelor of Technology in Electronics and communication is a
bonafide record of the mini project work carried out by them under our guidance and
supervision. This report in any form has not been submitted to any other University or Institute
for any purpose.

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DECLARATION

We, the undersigned, hereby declare that the mini project report entitled: **“SCRAMBLEBOX: An Arduino-Based Word Jumble Game Console”** submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology of APJ Abdul Kalam Technological University, Kerala, is a bonafide work carried out by us under the supervision of **Mr. KUMAR G S**, Assistant Professor, Department of Electronics and Communication Engineering. This submission represents our ideas in our own words, and where ideas or words of others have been included, we have adequately and accurately cited and referenced the original sources. We also declare that we have adhered to the ethics of academic honesty and integrity and have not misrepresented or fabricated any data, ideas, facts, or sources in our submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and may also invoke penal action from the original sources that have not been properly cited or from whom proper permission has not been obtained.

This report has not previously formed the basis for the award of any degree, diploma, or similar title from any other University.

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ABSTRACT

The ScrambleBox: Arduino-based Word Jumble Game is an interactive project that challenges players to unscramble words using an Arduino microcontroller. It features a user-friendly interface with buttons for input, a display for showing scrambled words, and scoring based on performance, making it an engaging educational tool for enhancing vocabulary skills. The ScrambleBox Word Jumble Game is an innovative educational project designed to enhance vocabulary and cognitive skills through interactive gameplay. Utilizing an Arduino microcontroller, the game presents players with scrambled words that they must decipher within a limited number of attempts.

Key Features include user interaction, A dynamic scoring system, visual and sound feedback and a unique word scrambling logic. This project not only serves as a fun activity but also as a practical application of programming and electronics, making it suitable for educational settings and hobbyist exploration.

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LIST OF ABBREVIATIONS

Serial No.	Abbreviation	Definition
1	AI	Artificial Intelligence
2	AWG	American Wire Gauge
3	DC	Direct Current
4	GPIO	General Purpose Input/Output
5	I2C	Inter-Integrated Circuit
6	IDE	Integrated Development Environment
7	LCD	Liquid Crystal Display
8	LED	Light Emitting Diode
9	OLED	Organic Light Emitting Diode
10	PCB	Printed Circuit Board
11	PWM	Pulse Width Modulation
12	RAM	Random Access Memory
13	SCL	Serial Clock Line - I2C
14	SDA	Serial Data Line - I2C
15	SRAM	Static RAM
16	TFT	Thin-Film Transistor
17	USB	Universal Serial Bus
18	VCC	Voltage Common Collector

CHAPTER 1

INTRODUCTION

Language is a fundamental aspect of human communication, and vocabulary plays a crucial role in effective expression and comprehension. Word jumble games provide an engaging and interactive way to enhance vocabulary, cognitive skills, and problem-solving abilities. The Scramblebox Word Jumble Game is an Arduino-based interactive console designed to challenge players in an educational and entertaining manner by unscrambling words. This project utilizes an Arduino microcontroller to manage the game's logic, a Liquid Crystal Display (LCD) to display scrambled words and game status, buttons for user interaction, and a scoring system to encourage repeated play and skill improvement. Players can select from different word categories and manipulate scrambled letters using an intuitive interface. The system provides real-time feedback, hints, and scoring to make the gameplay engaging and rewarding. The primary objective of this project is to develop a portable and user-friendly word jumble game that not only entertains but also improves language skills and logical thinking. By integrating embedded systems and game design, this project demonstrates the practical application of electronics, programming, and user interface development in an educational context. This report details the design, implementation, and working principles of the Scramblebox Word Jumble Game, highlighting its potential use in educational settings, hobbyist projects, and interactive learning environments.

1.1 BACKGROUND AND HISTORY

Word games have been a popular form of entertainment and education for centuries, helping individuals develop language skills, enhance cognitive abilities, and improve problem-solving techniques. From traditional anagram puzzles and crossword puzzles to modern digital word games, the concept of rearranging letters to form meaningful words has been widely utilized across different platforms and generations.

Evolution of Word Jumble Games:

The origins of word jumble games can be traced back to classical word puzzles that required players to rearrange letters to discover hidden words. Over time, these puzzles evolved into structured games, such as:

- **Anagrams** – Dating back to ancient civilizations, anagrams were used as linguistic exercises and cryptographic tools.
- **Scrabble (1938)** – A board game that challenged players to form words from randomly drawn letters.
- **Jumble® Puzzle (1954)** – One of the most well-known word jumble games, appearing in newspapers worldwide.
- **Digital Word Games (2000s - Present)** – With the advent of computers and smartphones, word jumble games became widely available as mobile and web-based applications, such as **Word Scramble, Word Cookies, and Wordle**.

The Role of Technology in Word Games:

The integration of electronics and programming into word-based games has significantly enhanced the user experience. The Arduino platform, with its ability to control hardware components like LCDs, buttons, and sensors, provides an opportunity to develop interactive and educational gaming consoles. By leveraging microcontrollers, developers can create engaging word games that incorporate real-time feedback, scoring systems, and user interactions, making learning more effective and enjoyable.

The Scramblebox Word Jumble Game Concept:

The Scramblebox project builds upon this rich history by bringing the word jumble game into a portable and interactive format. Unlike traditional paper-based or digital app versions, Scramblebox integrates hardware components such as an Arduino microcontroller, LCD display, push buttons, and a scoring mechanism to create an immersive experience. This system is designed to:

- Encourage learning through gameplay by making vocabulary-building fun.
- Utilize embedded systems to demonstrate applications of microcontroller projects.
- Enhance cognitive skills by challenging players to think critically and solve jumbled word puzzles efficiently.

By combining traditional word puzzle mechanics with modern embedded technology, the Scramblebox Word Jumble Game represents an innovative approach to interactive learning and gaming, appealing to students, hobbyists, and language enthusiasts alike.

1.2 OBJECTIVE

The primary objective of the Scramblebox – An Arduino-based Word Jumble Game Console is to design and develop an interactive and educational word puzzle game that enhances vocabulary, cognitive skills, and problem-solving abilities. This project aims to bridge technology and learning by integrating embedded systems with an engaging word jumble challenge. The specific objectives are:

1. Develop an Interactive Gaming Console:

- Utilize an Arduino microcontroller to control game functions and user interactions.
- Implement an LCD display to provide real-time feedback and display scrambled words.

2. Enhance User Engagement and Learning:

- Create an intuitive button-based interface for selecting and rearranging letters.
- Implement a category system to allow users to choose between different word groups.
- Introduce hints and a scoring mechanism to encourage learning through repetition and improvement.

3. Implement a Randomized Word Scrambling Algorithm:

- Develop a logic to randomly scramble words from predefined categories.
- Ensure each game round presents a unique challenge by avoiding repetition of words.

4. Provide a Scoring and Feedback System:

- Reward players based on performance, with higher points for correct answers without hints.
- Display the final score at the end of the game, encouraging users to improve skills.

5. Showcase Practical Applications of Embedded Systems:

- Demonstrate how hardware and software work together in an interactive project.
- Provide insights into Arduino programming, sensor inputs, and real-time processing.

By achieving these objectives, the Scramblebox Word Jumble Game serves as both a fun and educational tool, promoting language learning, problem-solving, and hands-on experience with microcontrollers.

1.3 MOTIVATION

Games have long been recognized as an effective tool for learning and skill development, making complex concepts more engaging and accessible. Word jumble puzzles, in particular, have been widely used to enhance vocabulary, cognitive skills, and problem-solving abilities. The Scramblebox Word Jumble Game was inspired by the idea of merging educational gaming with embedded systems technology to create an interactive and enjoyable learning experience. The key motivations behind this project are:

1. **Bridging Technology and Education:**

- With the increasing importance of STEM (Science, Technology, Engineering, and Mathematics) education, there is a growing need for projects that combine electronics and programming with real-world applications.
- This project serves as an educational tool that introduces learners to Arduino, programming, and embedded systems in a fun and interactive way.

2. **Encouraging Interactive Learning:**

- Traditional learning methods can sometimes be passive and less engaging.
- The Scramblebox provides an interactive approach to improving vocabulary and cognitive abilities by challenging players to think critically and solve puzzles.

3. **Exploring Practical Applications of Embedded Systems:**

- Developing an Arduino-based game console showcases the potential of microcontrollers in gaming and user interaction.
- It provides hands-on experience in hardware integration, sensor usage, and user interface design.

4. **Fostering Problem-Solving and Logical Thinking:**

- The game mechanics encourage users to analyze scrambled words, form patterns, and make logical decisions, reinforcing problem-solving skills.
- The scoring and hint system adds an element of strategy, motivating users to refine their approach with each attempt.

5. **Creating an Engaging and Portable Gaming System:**

- Unlike mobile or web-based word games, the Scramblebox offers a physical and standalone gaming experience.
- Its compact and self-contained design makes it a portable and replayable educational tool.

6. Inspiring Innovation in Embedded Projects:

- This project demonstrates how Arduino microcontrollers can be used creatively for entertainment and learning.
- It serves as a foundation for future interactive electronics projects, encouraging innovation and experimentation in embedded systems development.

The Scramblebox Word Jumble Game was developed with the vision of making learning enjoyable and interactive while showcasing the capabilities of embedded systems. By blending education, technology, and gaming, this project serves as a unique and engaging learning tool that promotes both language development and technical skills.

1.4 RELEVANCE

The Scramblebox – An Arduino-Based Word Jumble Game Console is highly relevant in multiple domains, including education, embedded systems development, and interactive learning technologies. This project demonstrates the practical application of microcontrollers in gaming, while also serving as an engaging learning tool for vocabulary enhancement.

1. Educational Relevance:

- The game promotes active learning by encouraging players to think critically and solve word puzzles.
- Enhances vocabulary, spelling skills, and cognitive abilities, making it a valuable tool for students and language learners.
- Supports STEM (Science, Technology, Engineering, and Mathematics) education by integrating programming, electronics, and user interaction into a fun project.

2. Relevance in Embedded Systems and Technology:

- Showcases how Arduino microcontrollers can be used to create an interactive gaming console.
- Demonstrates real-world applications of LCD displays, button controls, randomization algorithms, and scoring systems in embedded systems.
- Encourages exploration of human-machine interaction, where hardware and software work together to provide an engaging experience.

3. Relevance in Gaming and Entertainment:

- Word games have remained popular for decades, with digital versions such as Scrabble, Wordle, and Jumble gaining significant attention.
- Unlike mobile apps, the Scramblebox provides a tangible, hands-on gaming experience, making it unique and engaging.
- The physical interaction with buttons, LEDs, and an LCD display offers a different experience from traditional screen-based word games.

4. Practical and DIY Project Relevance:

- Serves as an excellent project for engineering students, hobbyists, and makers interested in microcontroller-based applications.
- Provides a foundation for further innovation, such as adding speech recognition, multiplayer functionality, or IoT connectivity in future iterations.
- Encourages the development of problem-solving and coding skills through hands-on experimentation.

5. Social and Cognitive Relevance:

- Word jumble games help improve memory, pattern recognition, and problem-solving skills, making them beneficial for people of all ages.
- Can be used as an educational tool in schools, language training centers, and workshops to promote interactive learning.
- Engaging gameplay mechanics encourage replayability, making learning more enjoyable and effective.

The Scramblebox Word Jumble Game is relevant in multiple disciplines, from education and gaming to embedded systems and cognitive development. By integrating Arduino technology with an interactive word puzzle, it serves as a fun, educational, and technically enriching project that benefits both students and hobbyists alike.

CHAPTER 2

LITERATURE SURVEY

1. Word Jumble Logic in Game Development:

Hari Wiguna's Word Jumble Logic provides an essential framework for implementing word scramble logic in embedded systems. His work demonstrates how string manipulation and randomization techniques can be used to scramble words while maintaining a fair level of difficulty. This reference has been instrumental in designing the ScrambleBox game, as it provides insights into how scrambled words can be generated programmatically and then verified for correctness.

2. Arduino-Based String Manipulation Techniques:

The Arduino Forum discussion on string manipulation explores various approaches to handling character arrays, string operations, and optimization techniques in memory-constrained microcontrollers. Given that Arduino lacks native support for dynamic memory allocation like in Python or Java, handling strings efficiently is a key challenge. The insights from this reference helped in refining the ScrambleBox logic, particularly in word storage, retrieval, and scrambling.

3. Role of Arduino in Embedded Systems for Learning Applications:

The Arduino Reference Library provides a detailed overview of the core functions available in Arduino programming. It covers important functions for handling input/output operations, interfacing with hardware components like push buttons, LCD displays, buzzers, and potentiometers, and executing game logic. These functions are crucial in developing interactive educational tools like ScrambleBox. The reference served as a guide for implementing hardware-based word learning games using low-cost microcontrollers.

4. Educational Computer Games for E-Learning:

Bae et al. explored how educational computer games can be designed using activity theory to improve e-learning effectiveness. The study emphasizes the importance of engagement, interaction, and motivation in learning-based games. The research findings highlight that

traditional digital learning tools often lack physical interaction, which can reduce engagement levels. This aligns with the ScrambleBox project's goal of creating a physical word game that enhances cognitive learning while minimizing digital distractions.

5. DIY and Maker Culture in Game Development:

Instructables and Hackster.io are online platforms that provide numerous DIY (Do It Yourself) tutorials on Arduino-based games and educational tools. These resources emphasize the importance of hands-on learning, where students and hobbyists build their own gaming consoles or educational devices. The ScrambleBox project benefits from this maker culture, as it encourages innovation, creativity, and problem-solving through hardware-based implementations.

6. Tangible User Interfaces for Learning:

Sitdhisanguan et al. examined the effectiveness of Tangible User Interfaces (TUI) in educational games for children, particularly those with learning disabilities like autism. Their research showed that physical interaction improves engagement, memory retention, and motor skills. The ScrambleBox project incorporates TUI elements, such as physical buttons, a rotary potentiometer, and a buzzer feedback system, to provide an interactive and engaging word-learning experience that differs from traditional digital screens.

CHAPTER 3

METHODOLOGY

3.1 FLOW CHART OF CODE

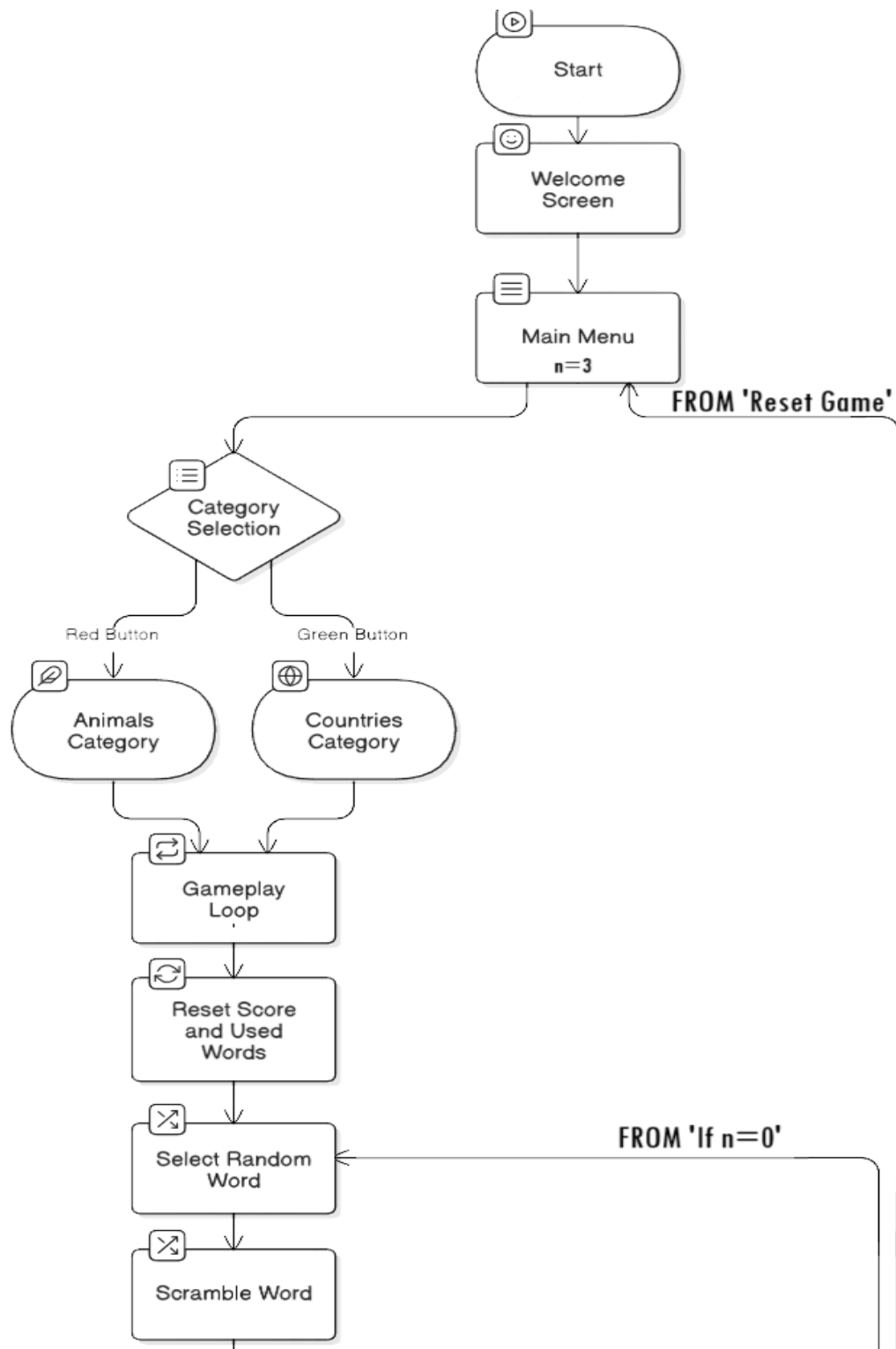


Figure 3.1.1 Code Flowchart Part 1

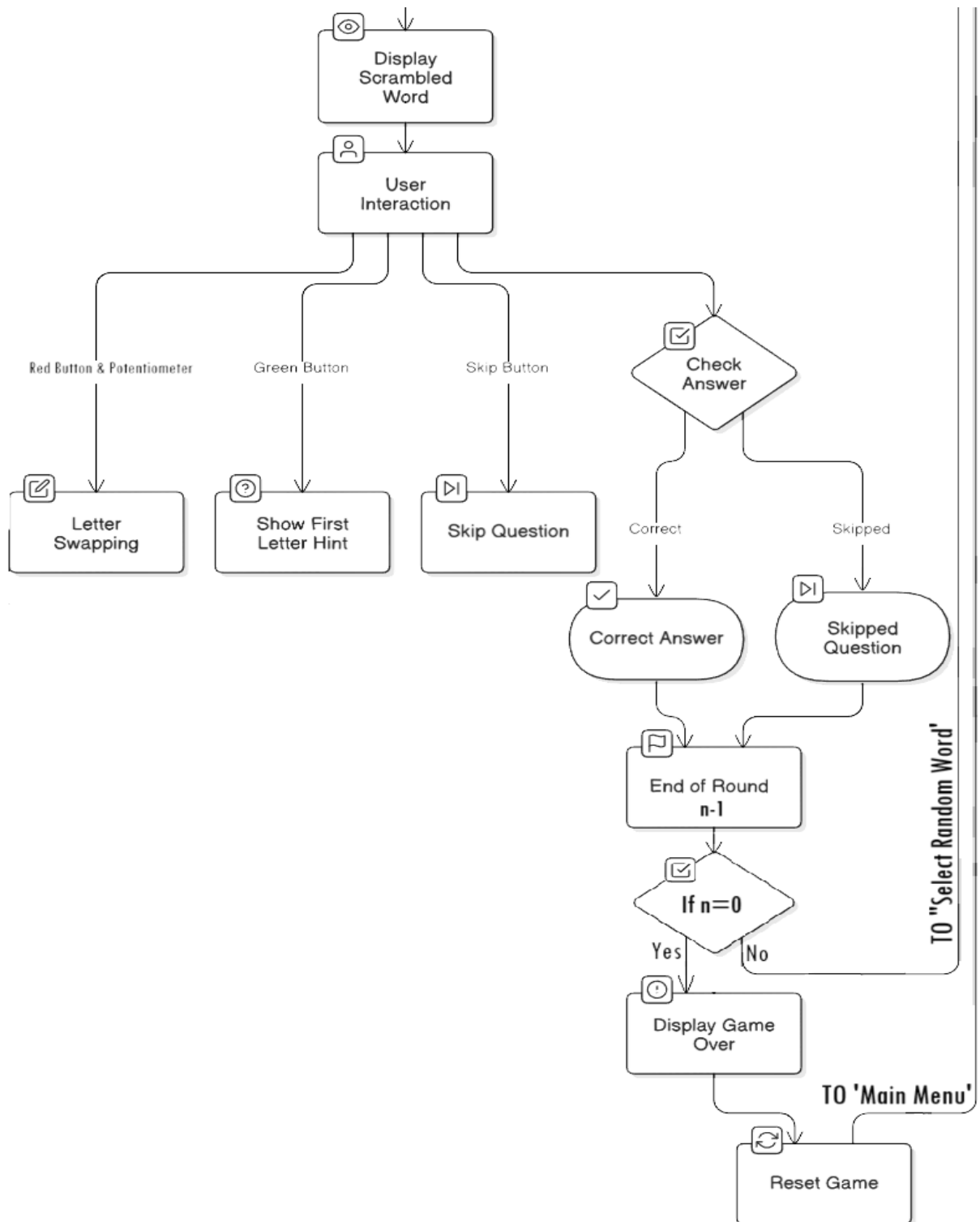


Figure 3.1.2 Code Flowchart Part 2

3.2 HARDWARE BLOCK DIAGRAM

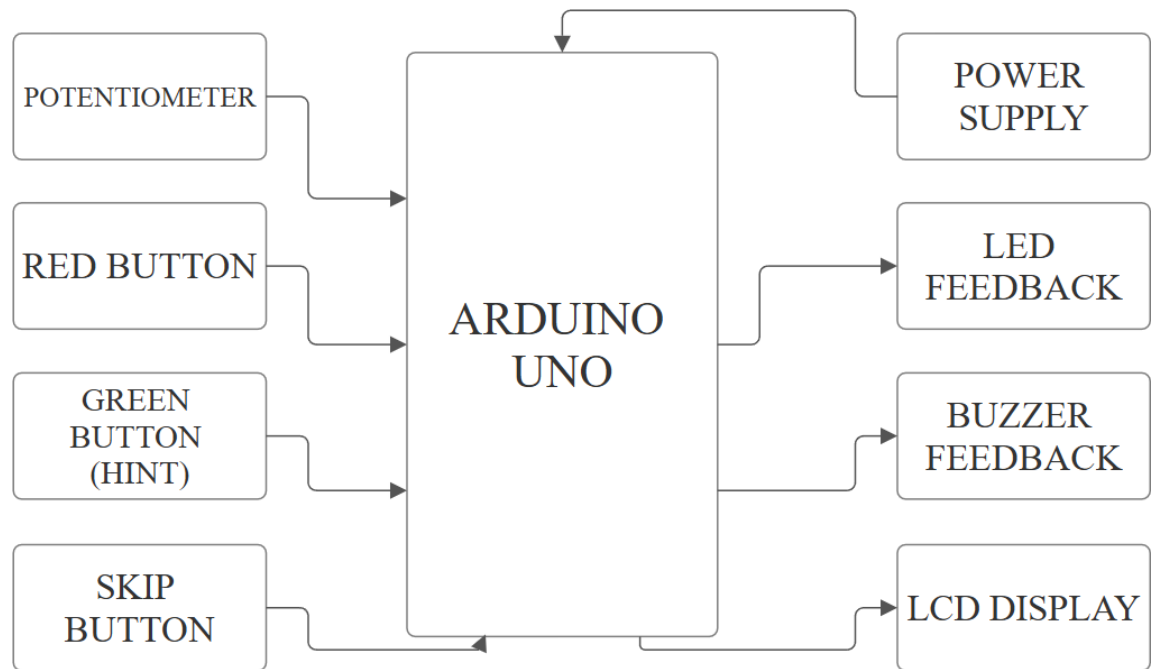


Figure 3.2 Hardware Block Diagram

3.3 WORKING PRINCIPLE

The Scramblebox Word Jumble Game is an interactive educational game designed to enhance vocabulary and cognitive skills through word unscrambling. The game operates using an Arduino microcontroller, which manages the game logic, user inputs, and outputs to the display and other components. The following sections outline the key components and their roles in the working principle of the game.

1. System Overview

The system consists of several hardware components, including an Arduino microcontroller, a 20x4 LCD with an I2C interface, buttons for user interaction, a potentiometer for cursor control, LEDs for feedback, and a buzzer for audio signals. The game is structured to present players with scrambled words that they must unscramble within a limited number of attempts.

2. Initialization

- Upon powering up, the Arduino initializes the LCD display and sets up the input and output pins for the buttons, LEDs, and buzzer.
- The game begins with a welcome screen displayed on the LCD, followed by a main menu prompting the user to start the game.

3. User Interaction

- The user interacts with the game using three buttons:
- **Red Button:** Used to select options and swap letters.
- **Green Button:** Used to request a hint (the first letter of the word).
- **Skip Button:** Used to skip the current question.
- A potentiometer is used to control the cursor position on the scrambled word displayed on the LCD.

4. Category Selection

- The user selects a category (Animals or Countries) by pressing the corresponding button.
- The game randomly selects a word from the chosen category that has not been used in previous rounds.

5. Word Scrambling

- The selected word is scrambled using a randomization algorithm, which rearranges the letters of the word.
- The scrambled word is displayed on the LCD, along with a cursor indicating the current letter position.

6. Gameplay Loop

- The user can manipulate the scrambled letters by moving the cursor with the potentiometer and swapping letters using the red button.
- If the user presses the green button, the first letter of the correct word is displayed as a hint.
- The user can skip the question by pressing the skip button, which moves the game to the next round.

7. Answer Checking

- After the user has rearranged the letters, the game checks if the scrambled word matches the correct word.
- If the user correctly unscrambles the word, the green LED lights up, and a success tone is played through the buzzer. The score is updated based on whether a hint was used.
- If the user skips the question, a message is displayed, and the game proceeds to the next round.

8. Scoring System

- The scoring system awards points based on the user's performance:
- 2 points for a correct answer without using a hint.
- 1 point for a correct answer after using a hint.
- The score is displayed at the end of the game.

9. Game Conclusion

- After all rounds are completed, the final score is displayed on the LCD, indicating the total points earned out of the maximum possible score.
- The game then resets, allowing the user to start a new game session.

The Scramblebox Word Jumble Game effectively combines hardware and software components to create an engaging and educational experience. By leveraging the capabilities of the Arduino microcontroller, the game provides a platform for users to enhance their vocabulary and cognitive skills in a fun and interactive manner. The project demonstrates the practical application of programming, electronics, and user interface design, making it suitable for educational settings and hobbyist exploration.

3.4 CIRCUIT DIAGRAM

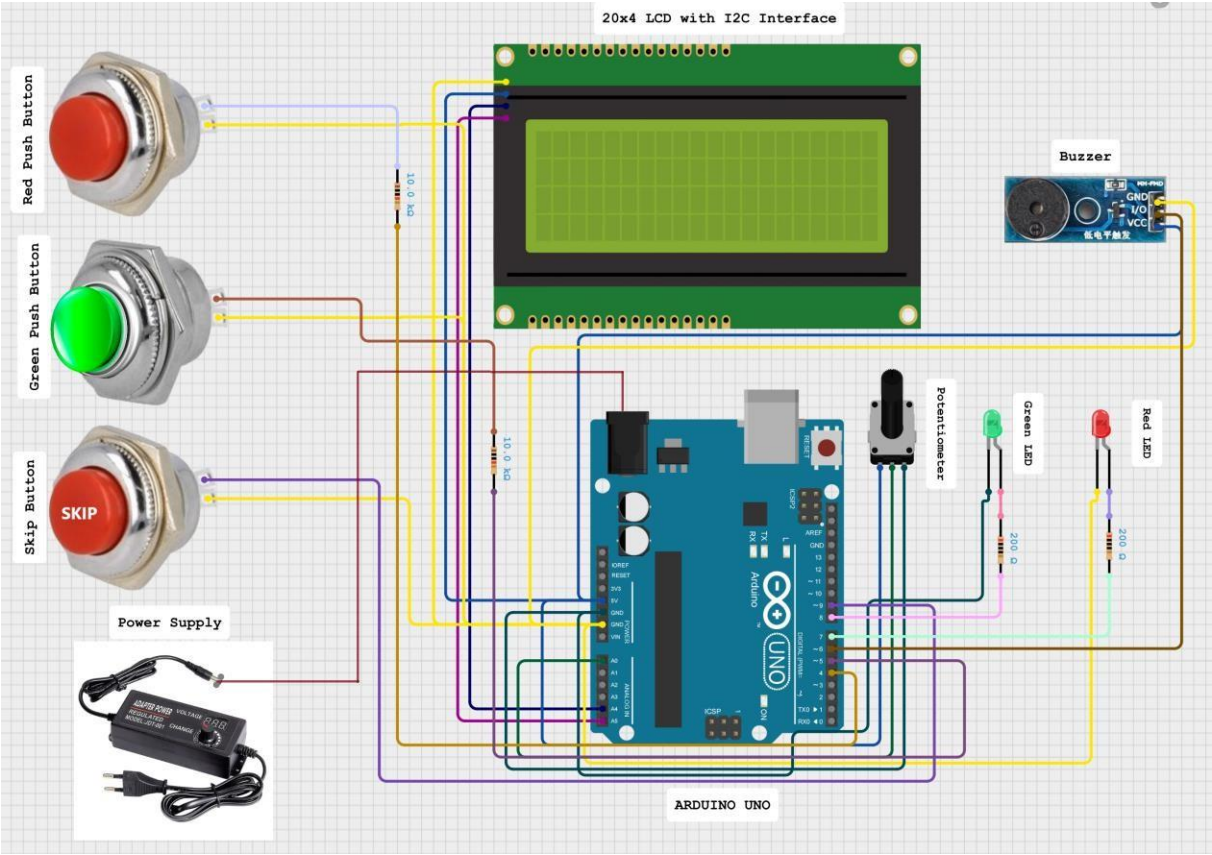


Figure 3.4 Circuit Diagram

CHAPTER 4

HARDWARE REQUIREMENTS

4.1 Power Source

- **Power Adapter (12V, 1A):**
 - **Purpose:** The 12V, 1A Power Adapter supplies stable power to the Arduino Uno and other components in the Scramblebox Word Jumble Game. It ensures consistent operation without relying on USB power, which may not provide sufficient current for all components.
 - **Explanation:**
 - The adapter provides a regulated 12V output that powers the Arduino Uno via the barrel jack.
 - The Arduino's onboard voltage regulator (7805 or AMS1117-5.0) converts this 12V to 5V, which is used to power the LCD, buttons, LEDs, and buzzer circuit.
 - The buzzer (CC1206AN) and other high-power components can be directly powered from the 12V supply using appropriate voltage regulation methods (e.g., a transistor switch for the buzzer).
 - **Considerations:**
 - **Voltage Compatibility:** The Arduino Uno supports 7V – 12V input, making a 12V, 1A adapter suitable.
 - **Current Requirements:**
 - Arduino Uno draws ~50mA – 100mA.
 - LCD, buttons, LEDs, and buzzer collectively consume less than 500mA, well within the adapter's 1A limit.
 - **Heat Dissipation:** Running 12V into Arduino's regulator generates some heat, but is manageable with proper ventilation.
 - **Polarity:** It follows the center-positive standard, ensuring correct polarity when connecting to the Arduino's DC jack.



Figure 4.1 Power Adapter

4.2 Development Board

- **Arduino UNO:**

- **Purpose:** The Arduino Uno serves as the main processing unit of the Scramblebox Word Jumble Game, handling user inputs, processing game logic, and controlling outputs such as the LCD display, LEDs, and buzzer. It facilitates real-time interactions between the user and the game system.
- **Explanation:** The Arduino Uno is an open-source microcontroller board based on the ATmega328P. It provides digital and analog I/O pins that allow interfacing with various electronic components. It runs on a simple programming environment (Arduino IDE) and can execute instructions to manage input-output operations effectively.
- **Considerations:**
 - **Power Supply:** The board operates on a 5V supply but can be powered using a 12V, 1A adapter via the barrel jack, regulated internally.
 - **Memory Constraints:** The ATmega328P has limited RAM (2KB), so efficient coding and memory management are crucial.
 - **Input/Output Limitations:** Since there are only limited pins, careful pin allocation is necessary for handling multiple inputs and outputs.
 - **Debouncing for Buttons:** Implementing debouncing in software prevents unintended multiple inputs from push buttons.

Pin Type	Pin Numbers	Description
Power Pins	3.3V, 5V, GND, VIN	Used for powering sensors and components. VIN allows external power input (7-12V).
Analog Input Pins	A0 - A5	Used to read analog signals (0-1023 range). Example: Potentiometer connected to A0.
Digital I/O Pins	0 - 13	Used for general input/output operations. Can be configured as input or output.
PWM Pins	3, 5, 6, 9, 10, 11	Support Pulse Width Modulation (PWM) for dimming LEDs or controlling motors.
Serial Communication Pins	0 (RX), 1 (TX)	Used for UART serial communication (USB or external modules).
SPI Pins	10 (SS), 11 (MOSI), 12 (MISO), 13 (SCK)	Used for high-speed communication with SPI devices (LCD, SD card modules, etc.).
I2C Pins	A4 (SDA), A5 (SCL)	Used for communication with I2C devices like the LCD display.
Reset Pin	RESET	Used to reset the microcontroller manually.

Table 4.2 Arduino UNO Pinout Table



Figure 4.2 Arduino UNO

4.3 Indicator Components

- **20x4 LCD (Liquid Crystal Display):**
 - **Purpose:** The 20x4 LCD is used to display scrambled words, hints, game instructions, user interactions, and the final score. It provides a clear and structured interface for players to engage with the Word Jumble Game.

- **Explanation:** A 20x4 LCD means it has 20 columns and 4 rows, allowing it to display up to 80 characters at a time. It operates using an HD44780 driver and communicates with the Arduino using either parallel (16-pin) or I2C (4-pin) interfaces. The I2C version reduces the number of required Arduino pins.
- **Considerations:**
 - **I2C vs. Parallel Connection:** Using an I2C adapter (SDA/SCL) simplifies wiring and saves digital pins.
 - **Power Requirements:** Operates at 5V, and backlight brightness may need adjustment using the potentiometer on the I2C module.
 - **Library Support:** Requires the LiquidCrystal_I2C library for easier communication in I2C mode.
 - **Contrast Adjustment:** A built-in potentiometer adjusts screen visibility for better readability.

Pin	Label	Description
1	VSS	Ground (0V)
2	VCC	Power Supply (5V)
3	V0	Contrast Adjustment (Potentiometer)
4	RS	Register Select (Command/Data)
5	RW	Read/Write Mode (Usually GND for Write)
6	E	Enable Signal
7-14	D0-D7	Data Pins (For 8-bit mode, all are used; for 4-bit mode, only D4-D7)
15	LED+	Backlight Positive (5V)
16	LED-	Backlight Ground (GND)

Table 4.3.1 LCD Pinout Table



Figure 4.3.1 20x4 LCD Display

- **Buzzer (CC1206AN):**

- **Purpose:** The CC1206AN Buzzer is used to provide audio feedback in the Scramblebox Word Jumble Game. It generates beeps to indicate correct and incorrect answers, button presses, and game start/end events, enhancing the interactive experience.

- **Explanation:**

- This buzzer has 3 pins: VCC, GND, and I/O (Signal).
- It operates on 3V – 12V DC, with 5V optimal for Arduino compatibility.
- The I/O pin is controlled via PWM signals from the Arduino, allowing frequency modulation to produce different tones.

- **Considerations:**

- **Operating Voltage:** 3V – 12V (Recommended: 5V).
- **Current Consumption:** Low (~20mA).
- **Sound Frequency:** Typically 2 kHz – 4 kHz, adjustable via PWM.
- **Avoiding Continuous Noise:** Use short beeps instead of a continuous tone.
- **Placement:** Keep away from sensitive components to minimize interference.

Pin	Label	Description	Arduino Connection
1	VCC	Power Supply (3V – 12V)	5V
2	I/O (SIG)	Signal Input (PWM)	Pin 6
3	GND	Ground (0V)	GND

Table 4.3.2 Buzzer Pinout Table



Figure 4.3.2 Buzzer

- **LEDs (Light Emitting Diodes):**

- **Purpose:** Serve as visual indicators to display the status of the circuit operation, such as indicating different brightness levels of the headlights.
- **Explanation:** Semiconductor devices that emit light when current flows through them in the forward direction.



Figure 4.3.3 LEDs

4.4 Passive Components

- **Resistors (2 x 330 Ω Resistors):**

- **Purpose:** Ensure proper operation of the circuit by limiting current flow and dividing voltages as needed.

- **Explanation:** Fixed resistors used for current limiting, voltage division, and biasing in different parts of the circuit.

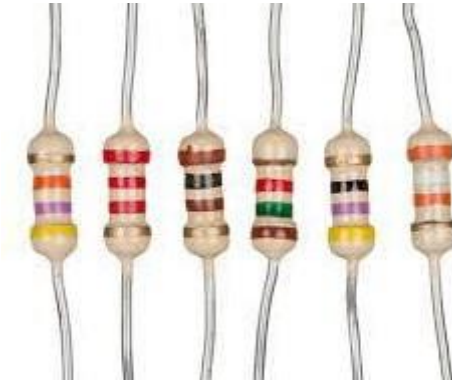


Figure 4.4.1 Resistors

- **Potentiometer (1 x 10k Ω Potentiometer):**

- **Purpose:** The potentiometer is used as an input device to control the selection of letters in the Scramblebox Word Jumble Game. It allows players to move the cursor over the scrambled word, enabling letter swapping and interaction with the game.
- **Explanation:** Variable resistors with three terminals allowing manual adjustment of resistance between two of its terminals.



Figure 4.4.2 Potentiometer

4.5 Input Devices

- **Momentary push buttons(3 x Push Buttons):**

- **Purpose:** Momentary push buttons are used as user input devices in the Scramblebox Word Jumble Game to interact with the system. They allow players to select options, confirm choices, swap letters, request hints, or skip words during gameplay.
- **Explanation:**
 - A momentary push button is a type of switch that remains ON only while it is being pressed and returns to OFF when released.
 - It consists of two pairs of contacts: Normally Open (NO) or Normally Closed (NC).
 - In this project, the buttons are used in a pull-down configuration (connected to GND with a pull-up resistor) to detect user presses as a HIGH signal (5V).
 - The buttons are connected to the Arduino digital input pins, allowing it to register user actions.
- **Considerations**
 - **Debouncing:** Since mechanical switches can cause bouncing effects, software **debouncing** is implemented to ensure stable input.
 - **Pull-up/Pull-down Resistors:** A 10k Ω pull-down resistor ensures the button reads LOW (0V) when not pressed.
 - **Placement:** Buttons should be easily accessible for intuitive user interaction.



Figure 4.5 Push Button

4.6 Interfacing Components

- **I2C Module**

- **Explanation:** A communication interface module that allows microcontrollers to connect with peripherals like LCD Module.
- **Purpose:** Reduces the number of GPIO pins required for communication, simplifying wiring and enabling efficient data transfer between components.



Figure 4.6 I2C Module

Pin Name	Pin Description
VCC	5V
GND	GND
SDA	D1
SCL	D2

Table 4.6 I2C Module Pinout Table

4.7 Circuit Assembly Components

- **Breadboard:**

- **Explanation:** A solderless prototyping board with interconnected holes that allow easy placement and connection of electronic components.
- **Purpose:** Enables quick circuit assembly and testing without permanent soldering, making it ideal for prototyping and debugging.

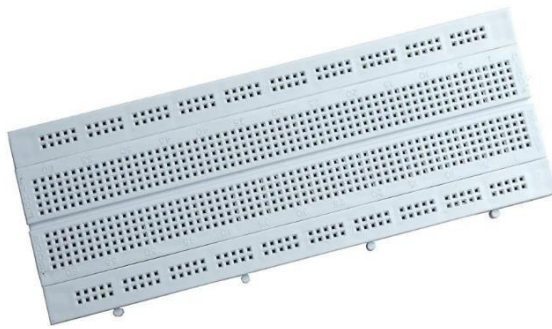


Figure 4.7.1 Breadboard

- **Jumper Wires:**

- **Explanation:** Insulated connecting wires with male or female connectors used to establish electrical connections between components on a breadboard or PCB.
- **Purpose:** Enables easy and flexible circuit assembly by connecting microcontrollers, sensors, and other electronic components without soldering.



Figure 4.7.2 Jumper Wires

- **Hookup wires:**
 - **Purpose:** Hookup wires are used to establish electrical connections between various components in the Scramblebox Word Jumble Game, allowing signals and power to be transmitted efficiently.
 - **Explanation**
 - Hookup wires act as the physical medium for electrical connections in the circuit.
 - They are used to connect the Arduino, LCD, buttons, LEDs, buzzer, potentiometer, and power supply on a breadboard or directly to components.
 - Different types of wires are used based on the connection type:
 - Male-to-Male: For connecting components on a breadboard.
 - Male-to-Female: For connecting to Arduino header pins.
 - Female-to-Female: For direct module-to-module connections.
 - **Considerations**
 - **Wire Gauge:** Typically 22 AWG or 24 AWG is used for low-power circuits.
 - **Length:** Should be long enough for easy wiring but not excessive to avoid clutter.
 - **Insulation:** Ensure proper insulation to prevent short circuits.
 - **Color Coding:**
 - Red – Power (VCC/5V/12V)
 - Black – Ground (GND)
 - Yellow/Blue/Green – Signal/Data Lines



Figure 4.7.3 Hookup wires

4.8 Tools and Testing Equipment

- **Soldering Iron and Solder:**

- **Purpose:** Essential for assembling the circuit if moving beyond the prototyping stage on a breadboard.
- **Explanation:** Tools used for soldering components to make permanent connections



Figure 4.8.1 Soldering iron and solder

- **Wire Cutters and Strippers:**

- **Purpose:** Facilitates neat and precise wiring, ensuring reliable electrical connections.
- **Explanation:** Tools used to cut and strip insulation from wires to appropriate lengths for connections.



Figure 4.8.2 Wire cutter and stripper

- **Multimeter:**

- **Explanation:** An electronic tool used to measure voltage, current, and resistance in electrical circuits.
- **Purpose:** Essential for testing circuit parameters, diagnosing faults, and ensuring proper operation during assembly and troubleshooting phases.



Figure 4.8.3 Multimeter

4.9 Structural Components

- **Forex Sheet:**

- **Explanation:** A sturdy sheet material used as the structural framework for mounting and securing electronic components in the system.
- **Purpose:** Provides a stable base for assembling the Smart Pill Dispensing system, ensuring durability and proper organization of components.



Figure 4.9 Forex Sheets

CHAPTER 5

RESULTS



Figure 5.1 Prototype

The Scramblebox Word Jumble Game was successfully implemented and tested, demonstrating its functionality in real-world scenarios. Below are the key outcomes:

1. System Functionality

- The Arduino Uno effectively managed the game logic, handling user inputs and controlling the LCD display, LEDs, and buzzer.
- The 20x4 LCD with I2C provided clear visibility for scrambled words, hints, and score updates.
- The button interface allowed seamless user interaction, including word selection, letter swapping, and hints.
- The potentiometer-based cursor control enabled precise letter selection and swapping.

2. Game Performance

- Words were successfully scrambled using a randomization algorithm, ensuring a unique challenge in each round.
- The scoring system correctly awarded points based on performance, encouraging user engagement.
- The game logic ensured that words were not repeated within a session, maintaining variety.

3. User Experience

- Players were able to navigate the game with ease due to the intuitive button and display system.
- The LED feedback and buzzer alerts provided real-time confirmation of correct answers and actions.
- The hint system effectively assisted players in progressing through the game.

4. Reliability and Stability

- The game operated consistently without crashes, successfully resetting after completion.
- The hardware components (buttons, LCD, potentiometer, and buzzer) functioned as expected under multiple test conditions.
- Power supply (12V, 1A adapter) provided stable operation without overheating or voltage drops.

The Scramblebox Word Jumble Game successfully met its objectives, providing an engaging and educational experience. The combination of hardware and software elements resulted in a smooth, interactive word puzzle game that can be used for vocabulary development, cognitive skill improvement, and entertainment. Future enhancements could include additional word categories, difficulty levels, and multiplayer functionality

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

The Scramblebox Word Jumble Game successfully merges hardware and software components to create an engaging, interactive, and educational experience for users. By utilizing an Arduino Uno, a 20x4 LCD with an I2C interface, buttons for user interaction, a potentiometer for cursor control, LEDs for feedback, and a buzzer for audio signals, the system provides an intuitive and dynamic gameplay experience.

Throughout the development and testing phases, the game demonstrated stable performance and accurate responsiveness in handling user inputs and displaying real-time feedback. The randomized word scrambling algorithm ensured that each round was unique, maintaining the challenge and engagement levels for users. The scoring system further motivated users by rewarding them based on their accuracy and use of hints, making the game an effective tool for improving vocabulary and cognitive skills.

One of the key achievements of this project was the successful implementation of user-friendly controls through a button-based selection system and a potentiometer-controlled cursor movement. This ensured a smooth and immersive gameplay experience while reinforcing concepts of real-time user input processing in embedded systems. The integration of LEDs and a buzzer provided instant visual and auditory feedback, enhancing the overall user experience. From a technical standpoint, the project effectively utilized fundamental principles of embedded systems, electronics, and programming. The use of an I2C-based LCD reduced the number of required connections, optimizing the wiring and improving the overall system efficiency. The button debouncing mechanisms and structured power supply (12V, 1A adapter) ensured consistent and reliable operation without any unintended disruptions or overheating issues.

Moreover, the project underscores the potential of Arduino-based educational games in providing interactive learning experiences. The Scramblebox Word Jumble Game is not only a fun recreational tool but also an excellent demonstration of programming logic, hardware interfacing, and real-time decision-making in embedded systems.

Future Scope and Improvements:

While the project successfully meets its objectives, there are several areas for future enhancements:

1. **Expanded Word Database** – Including more word categories and difficulty levels for a more diverse challenge.
2. **Multiplayer Mode** – Allowing two or more players to compete against each other.
3. **Wireless Connectivity** – Integrating Bluetooth or Wi-Fi to enable online gameplay or remote monitoring.
4. **Advanced Display Features** – Incorporating graphical elements on an OLED or TFT display for an improved user interface.
5. **AI-Based Adaptive Difficulty** – Implementing an AI system to adjust the difficulty based on the player's performance.

The Scramblebox Word Jumble Game stands as a testament to the innovative potential of embedded systems in education and entertainment. By bridging the gap between technology and learning, this project highlights the practical applications of microcontroller-based systems in interactive games. It serves as an excellent foundation for future research and development in the fields of Arduino-based gaming, interactive learning tools, and smart educational devices. This project not only showcases the capabilities of embedded systems but also encourages further exploration into intelligent and interactive gaming solutions. Through continuous improvements and technological advancements, similar projects could significantly contribute to innovative learning experiences in the future.

This is a much more detailed and structured conclusion that ties together the achievements, technical aspects, user experience, and future scope of the project.

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APPENDIX 1

ARDUINO CODE:

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <string.h>

// ----- LCD Setup -----
LiquidCrystal_I2C lcd(0x27, 20, 4); // 20x4 LCD at I2C address 0x27

// ----- Pin Definitions -----
#define RED_BUTTON 4 // Menu selection/letter swapping
#define GREEN_BUTTON 5 // Hints
#define SKIP_BUTTON 9 // Question skipping
#define POT_PIN A0 // Potentiometer
#define BUZZER 6 // Buzzer
#define RED_LED 7 // Wrong answer
#define GREEN_LED 8 // Correct answer

// ----- Game Constants -----
const int TOTAL_ROUNDS = 3;

// ----- Word Banks -----
char animals[][6] = {"TIGER", "ZEBRA", "SHARK", "HORSE", "EAGLE", "SNAKE",
"WHALE", "OTTER", "ROBIN", "PANDA"};
char countries[][6] = {"INDIA", "CHILE", "EGYPT", "QATAR", "ITALY", "JAPAN",
"YEMEN", "SUDAN", "SPAIN", "GHANA"};

// ----- Global Variables -----
int category = -1;
char correctWord[6];
char scrambledWord[6];
int cursorPos = 0;
int selectedPos = -1;
bool hintUsed = false;
int score = 0;
int roundNum = 0;
bool usedWords[10] = {false}; // Track used words (10 is the max number of words in each
category)
```

```

// ----- Function Prototypes -----
void showWelcomeScreen();
void showMenu();
void chooseCategory();
void playGame();
void updateDisplay();
void handleUserInput();
void moveLetter(int from, int to);
void scrambleWord(const char* input, char* output);
void correctAnswer();
void skipQuestion();
void showFinalScore();
void resetUsedWords(); // Function to reset used words

// ----- Setup -----
void setup() {
    lcd.init();
    lcd.backlight();

    pinMode(RED_BUTTON, INPUT_PULLUP);
    pinMode(GREEN_BUTTON, INPUT_PULLUP);
    pinMode(SKIP_BUTTON, INPUT_PULLUP);
    pinMode(BUZZER, OUTPUT);
    pinMode(RED_LED, OUTPUT);
    pinMode(GREEN_LED, OUTPUT);

    randomSeed(analogRead(A1));
    showWelcomeScreen();
}

void loop() {
    showMenu();
}

// ----- Welcome Screen -----
void showWelcomeScreen() {
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("WELCOME TO");
    lcd.setCursor(0, 2);
    lcd.print("SCRAMBLEBOX");
    delay(2000);
}

```

```

// ----- Simplified Main Menu -----
void showMenu() {
  lcd.clear();
  lcd.print("PRESS RED BUTTON");
  lcd.setCursor(0, 2);
  lcd.print("TO START GAME");

  while (true) {
    if (digitalRead(RED_BUTTON) == LOW) {
      delay(300);
      chooseCategory();
      break;
    }
  }
}

// ----- Category Selection -----
void chooseCategory() {
  lcd.clear();
  lcd.print("CATEGORIES:");
  lcd.setCursor(0, 2);
  lcd.print("ANIMALS (Red)");
  lcd.setCursor(0, 3);
  lcd.print("COUNTRIES (Green)");

  while (true) {
    if (digitalRead(RED_BUTTON) == LOW) {
      category = 0;
      lcd.clear();
      lcd.print("ANIMALS SELECTED");
      delay(2000);
      playGame();
      break;
    }
    if (digitalRead(GREEN_BUTTON) == LOW) {
      category = 1;
      lcd.clear();
      lcd.print("COUNTRIES SELECTED");
      delay(2000);
      playGame();
      break;
    }
  }
}

```

```

// ----- Reset Used Words -----
void resetUsedWords() {
    for (int i = 0; i < 10; i++) {
        usedWords[i] = false; // Reset all used words
    }
}

// ----- Gameplay -----
void playGame() {
    score = 0;
    resetUsedWords(); // Reset used words at the start of the game

    for (roundNum = 0; roundNum < TOTAL_ROUNDS; roundNum++) {
        int idx;
        bool wordSelected = false;

        // Select a random word that hasn't been used
        while (!wordSelected) {
            idx = random(0, category == 0 ? sizeof(animals) / sizeof(animals[0]) :
sizeof(countries) / sizeof(countries[0]));
            if (!usedWords[idx]) {
                usedWords[idx] = true; // Mark this word as used
                wordSelected = true; // A valid word has been selected
            }
        }

        // Copy the selected word and scramble it
        strcpy(correctWord, category == 0 ? animals[idx] : countries[idx]);
        scrambleWord(correctWord, scrambledWord);

        cursorPos = 0;
        selectedPos = -1;
        hintUsed = false;

        while (true) {
            updateDisplay();
            handleUserInput();

            if (digitalRead(SKIP_BUTTON) == LOW) {
                skipQuestion();
                break;
            }
        }
    }
}

```

```

    if (strcmp(scrambledWord, correctWord) == 0) {
        correctAnswer();
        delay(1000);
        break;
    }
}
}
showFinalScore();
}

// ----- Improved Cursor Handling -----
void updateDisplay() {
    lcd.clear();
    lcd.print(scrambledWord);
    // Ensure cursor stays within word bounds
    cursorPos = constrain(cursorPos, 0, strlen(scrambledWord)); // Adjusted to allow cursor to
reach the last position
    lcd.setCursor(cursorPos, 1);
    lcd.print("^");
}

void handleUserInput() {
    static int lastPotValue = -1; // Store the last potentiometer value
    int potVal = analogRead(POT_PIN);

    // Map the potentiometer value to the range of the scrambled word length
    cursorPos = map(potVal, 0, 1023, 0, strlen(scrambledWord)); // Ensure it maps to the full
length

    // Only update cursor position if the potentiometer value has changed significantly
    if (abs(potVal - lastPotValue) > 10) { // Adjust the threshold as needed
        cursorPos = constrain(cursorPos, 0, strlen(scrambledWord)); // Adjusted to allow cursor to
reach the last position
        lastPotValue = potVal; // Update the last potentiometer value
    }

    if (digitalRead(RED_BUTTON) == LOW) {
        delay(300);
        if (selectedPos == -1) {
            selectedPos = cursorPos;
        } else if (cursorPos != selectedPos) {
            moveLetter(selectedPos, cursorPos);
            selectedPos = -1;
        }
    }
}

```

```

    }
}

if (digitalRead(GREEN_BUTTON) == LOW && !hintUsed) {
    hintUsed = true;
    lcd.setCursor(0, 2);
    lcd.print("FIRST LETTER: ");
    lcd.print(correctWord[0]);
    delay(2000);
}
}

// ----- Letter Movement -----
void moveLetter(int from, int to) {
    char temp = scrambledWord[from];
    if (from < to) {
        for (int i = from; i < to; i++) {
            scrambledWord[i] = scrambledWord[i + 1];
        }
    } else {
        for (int i = from; i > to; i--) {
            scrambledWord[i] = scrambledWord[i - 1];
        }
    }
    scrambledWord[to] = temp;
}

// ----- Word Scrambling -----
void scrambleWord(const char* input, char* output) {
    strcpy(output, input);
    for (int i = 0; i < strlen(output); i++) {
        int j = random(strlen(output));
        char temp = output[i];
        output[i] = output[j];
        output[j] = temp;
    }
}

// ----- Skip Functionality -----
void skipQuestion() {
    digitalWrite(RED_LED, HIGH);
    tone(BUZZER, 500, 300);
    delay(300);
    noTone(BUZZER);
}

```

```

digitalWrite(RED_LED, LOW);
lcd.clear();
lcd.print("QUESTION SKIPPED!");
delay(2000);
}

// ----- Correct Answer -----
void correctAnswer() {
    digitalWrite(GREEN_LED, HIGH);
    tone(BUZZER, 1000, 1000);
    score += hintUsed ? 1 : 2; // Score based on whether a hint was used
    delay(1000);
    digitalWrite(GREEN_LED, LOW);

    // Show the correct word for a few seconds
    lcd.clear();
    lcd.print("CORRECT! The word:");
    lcd.setCursor(0, 2);
    lcd.print(correctWord);
    delay(3000); // Display the correct word for 3 seconds
}

// ----- Final Score Display -----
void showFinalScore() {
    lcd.clear();
    lcd.print("GAME OVER!");
    lcd.setCursor(0, 1);
    lcd.print("SCORE: ");
    lcd.print(score);
    lcd.print(" OUT OF 6"); // Adjusted to reflect the total possible score
    delay(5000);
    showMenu();
}

```

APPENDIX 2

LIST OF COMPONENTS:

Sl. No.	Components	Cost	Quantity
1	Arduino UNO	₹498	1
2	Push Buttons, LEDs	₹40	3, 2
3	Buzzer	₹44	1
4	LCD + I2C module	₹408	1
5	Resistors	₹1	2
6	Jumper Wires	₹35	15
7	Power Adapter 12V 1A	₹180	1
8.	Forex Sheet	₹60	40cm x 40cm
9	Bread Board	₹50	1
10.	Other Miscellaneous Materials	₹30	-
11.	Hookup Wires	₹20	10
12.	Potentiometer	₹20	1
Total		₹1386	